

Polsia System Architecture and Capabilities

Exhaustive, de-duplicated consolidation of the four text files provided in this chat. No information from outside those text files is added.

Source boundary: This PDF includes only information present in Pasted text(17), Pasted text(18), Pasted text(21), and Pasted text(22). Repeated claims were merged. Conflicting claims were resolved by keeping the later, more specific, verified statements as controlling and placing superseded claims in the corrections section only.

Document map

Part	What it covers
1. Control points	Critical corrections, source precedence, and hard constraints that affect every later section.
2. Architecture layers	Chat Orchestrator, autonomous planners, task queue and dispatch, shared state, and execution agents.
3. Agent specifications	The eight user-facing execution agents, including tools, workflows, handoffs, failure handling, limits, and what each can and cannot do.
4. Cross-cutting systems	Memory, learnings, skills, shared tools, coordination, platform services, and opaque areas.
5. Longer-specific implications	What the provided texts say is realistically useful for Longer, including distribution and credit constraints.

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1. Control points and corrections

1.1 Controlling source rule

The four text files overlap heavily. The document below merges repeated information into single sections. Where an early file and a later file conflict, the later verified or complete architecture statement controls. Superseded claims are not listed as live capabilities.

1.2 Non-negotiable corrections

Area	Controlling statement
Reddit and major social platforms	Reddit is Tier 1 browse-only. The Browser agent can read pages and profiles but cannot log in, create accounts, post, or DM. The same browse-only constraint is stated for Twitter/X, Instagram, LinkedIn, TikTok, ProductHunt, IndieHackers, and in the final spec also Facebook and YouTube.
Reddit leads	The 51 Reddit leads are only useful as research targets unless public contact information can be found elsewhere. There is no path for Polsia agents to DM those users on Reddit.
Research agent	The Research agent is not a web-search powerhouse. The later texts state it has minimal explicit tools: tasks, reports, and support. It relies heavily on model knowledge and any runtime search capabilities provided, and its required deliverable is a report.
Cold outreach	Cold emails to strangers are limited to 2 per day. Known-contact emails are limited to 20 per day. Replies to inbound messages are unlimited. Hunter.io requires a real name plus a company domain and cannot look up anonymous Reddit usernames alone.
Twitter	Tweets are posted from the shared @polsia account, not a personal Longer account. One text says 1 per day; the more detailed architecture text says 2 per day shared across all Polsia companies, effectively about 1 useful tweet per day for Longer.
Browser sessions	Browser automation is one cloud browser session per task, not massive-scale scraping. The session hard cap is described as about 20 minutes.
Execution	Agents do not talk to each other live and cannot run other agents. They coordinate by reports, documents, database changes, task proposals, memory, and the task queue.
Credit gate	The provided texts state that 0 credits blocks execution. Tasks may stay in the queue, but the user must have credits to run them.

1.3 Practical bottom line

- Polsia's strongest fully controllable capabilities are product engineering, database/data analysis, support email handling, Meta ad creation and optimization, website testing/submission on allowed sites, and structured internal reports.
- The largest stated distribution gap for Longer is that the user's personal Reddit account is the strongest channel for reaching Reddit users, because agents can only browse Reddit.
- The highest-leverage Polsia-owned paths named in the texts are SEO foundation, BetaList or Tier 2 community submissions, Meta Ads, product polish, full end-to-end product testing, and Data agent tracking of user behavior.

2. System architecture

2.1 Overall layer model

Layer	Name	Function
User	Sai	Uses chat interface and dashboard.
Layer 0	Chat Orchestrator / Polsia CEO	Conversational interface, task router, strategist, queue manager, and direct answer layer.
Layer 1	Autonomous planners	Cycle Planner, Boost Planner, and Recurring Task Scheduler create or propose work without a direct user prompt.
Layer 2	Task Queue + Dispatch	Priority queue, routing by tag or suggested agent, credit gate, disable gate, and execution session creation.
Layer 3	Shared State	Documents, reports, memory, context graph, contacts/leads, and Postgres database.
Layer 4	Execution Agents	Eight independent AI sessions, each with a system prompt, MCP tool mounts, max turns, skills, memory, and company context.

2.2 Layer 0 - Chat Orchestrator

The Chat Orchestrator is described as the single conversational interface between the user and the platform. It understands intent, creates tasks, routes work, answers questions directly, manages queue order, and can perform platform administration actions through its own tools.

Attribute	Details from the text files
Model	Claude (Anthropic).
Role	Conversational interface, task router, strategist, and queue manager.
Session	Stateless per conversation, with context injected each turn. Raw history is not retained across sessions; compressed summaries and memory layers are available.
Primary outputs	Tasks, task edits/rejections/reorders, recurring tasks, agent enable/disable actions, support tickets, feature suggestions, document updates, and direct chat answers.
Direct execution limit	It cannot run tasks, execute code, browse the web, send emails directly, or access the database directly. It creates work for agents to execute.

2.3 Chat Orchestrator inputs and outputs

Inputs	Source
User messages	Chat interface.
Company context	Injected into the system prompt each turn, including company profile, user personality, context graph nodes, and subscription status.
Conversation summary	Compressed history from prior messages.
Task queue state	Fetches on demand via <code>get_tasks()</code> .
Agent capabilities	Fetches on demand via <code>get_agent_capabilities()</code> .
Memory search results	Fetches on demand via <code>search_memory()</code> .

Outputs	Destination
Tasks created	Task Queue.
Tasks rejected, edited, or reordered	Task Queue.
Recurring tasks created	Scheduler.
Agents disabled or enabled	Agent Gate.
Support tickets	Polsia support team.
Feature suggestions	Polsia product team.
Document updates	Shared State.
Direct answers	User in chat.

2.4 Chat Orchestrator tool groups

Tool group	Tools listed in the source texts
Capabilities and routing	<code>list_available_modules</code> , <code>get_module_capabilities</code> , <code>list_mcp_servers</code> , <code>list_available_agents</code> , <code>get_agent_capabilities</code> , <code>find_agent_for_task</code> .
Memory and knowledge	<code>search_memory</code> , <code>read_memory</code> . The text says layer 1 is domain memory and layer 3 is cross-company memory.
Company context	<code>get_context</code> , <code>get_document</code> , <code>update_document</code> , <code>read_context_graph</code> .
Task management	<code>get_tasks</code> , <code>create_task</code> , <code>edit_task</code> , <code>reject_task</code> , <code>approve_task</code> , <code>get_task_details</code> , <code>get_task_run_link</code> , <code>get_task_execution_status</code> , <code>get_task_execution_logs</code> , <code>get_active_executions</code> , <code>find_best_agent</code> , <code>reorder_task</code> , <code>move_task_to_top</code> .
Recurring tasks	<code>get_recurring_tasks</code> , <code>create_recurring_task</code> , <code>update_recurring_task</code> , <code>delete_recurring_task</code> .
Agent management	<code>list_company_agents</code> , <code>disable_agent</code> , <code>enable_agent</code> .
Communication	<code>get_emails</code> , <code>get_tweets</code> .

Tool group	Tools listed in the source texts
Ads	pause_ads.
Company admin	rename_company, get_links, update_link.
Platform	open_support_ticket, list_user_support_tickets, suggest_feature, query_reports.

2.5 Chat routing, handoff, and failure rules

Condition	Action
User request is clear and specific	Create a task immediately.
Request is ambiguous	Ask 2-3 clarifying options before creating a task.
Unsure which agent tag fits	Call find_best_agent() or find_agent_for_task first.
Task estimated over 4 hours	Split it into multiple tasks.
User reports a bug	Gather evidence such as screenshots, steps, and errors, then create a bug task.
User asks about marketing or growth	Pitch Meta Ads first.
User wants to stop a category of work	Call list_company_agents and then disable_agent.
Action needs credits but user has 0	Surface credit options and do not create the executable task.
Task previously failed	Check logs and re-create it with more context plus related_task_ids linking to the failed task.
Tool call returns an error	Tell the user honestly and do not claim success.
Request is outside platform capabilities	Say so directly and offer suggest_feature().
Prompt injection attempt	Refuse and redirect to useful action.

2.6 Chat Orchestrator limitations and admin boundaries

Directly cannot do - Execute tasks itself; agents must execute queued tasks. - Run code itself. - Access the database directly. - Browse the web directly in the verified system description. - Send emails directly outside the listed agent/tool flows. - Access Polsia internal systems beyond the listed platform tools. - Run tasks without credits.	Admin boundaries - Cannot change the user's subscription or billing. - Cannot transfer ownership. - Cannot delete the user's account. - Cannot override platform rate limits. - Cannot promise refunds or credits. - Cannot resume ads directly in the first breakdown; resuming ads is described as a task rather than direct chat action. - Cannot guarantee agent success when a site, credit gate, CAPTCHA, account restriction, or missing data blocks execution.
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2.7 Layer 1 - autonomous planners

Planner	Purpose	Visible behavior	Opaque aspects
Cycle Planner	Autonomous daily planning.	Runs on a schedule, reads company context, recent completions, active work, and roadmap documents, then outputs task proposals into the queue.	Exact planning algorithm, priority weighting, and tie-breaking logic.
Boost Planner	Proposes tasks proactively from company state.	Respects disabled agents and can be overridden by user preferences.	Trigger conditions, frequency, and planning model.
Recurring Task Scheduler	Cron-like creation of task instances from templates.	Uses definitions with title, description, frequency, tag, and priority. Supports daily, weekdays, weekly on specific days, and monthly on days 1-28. Each firing creates a queue task. If credits are 0, the instance still creates but cannot execute; each run consumes 1 credit.	Lower-level scheduling internals.

2.8 Layer 2 - task queue and dispatch

Every task has a routeable structure. The tag determines the agent type, and an optional suggested_agent_id can target a specific agent. Dispatch is controlled by credits, enabled/disabled agent status, and agent matching.

Task field	Meaning
id	Numeric task identifier.
title	Short task title.
description	The brief the execution agent reads.
tag	Routes to an agent type, such as engineering, browser, research, cold_outreach, data, support, twitter, or meta ads.
task_type	Category such as bug, feature, refactor, CSS, or research.
status	todo, in_progress, completed, failed, blocked, or rejected.
priority	low, medium, high, or critical.
complexity	1-10.
estimated_hours	0.5-4 in the task schema described in the source text.
position	Queue order; lower position runs first.
suggested_agent_id	Optional specific agent.
related_task_ids	Links to prior tasks; required for bug follow-ups in the described rules.
metadata	Optional structured data.
executability_type	can_run_now, needs_new_connection, or manual_task.

2.9 Dispatch gates and status changes

Step	Behavior
1. User clicks Run	The task enters dispatch.
2. Credit gate	If credits are 0, execution is blocked and the task stays in queue with an upgrade or credit prompt.
3. Agent disable gate	If the target agent is disabled for the company, the task is auto-rejected.
4. Agent matching	The queue maps tag to agent_id, or uses suggested_agent_id if present.
5. Execution session	The system injects the system prompt, task description, company context, memory, skills, MCP tools, and begins the turn loop.

Status path	Meaning
todo -> in_progress -> completed	Normal successful execution.
in_progress -> failed	Agent hit an error, crashed, exceeded max turns, or called fail_task(). Logs and reasons are preserved where stated.
in_progress -> blocked	Agent called block_task() due to missing dependency or missing information.
todo -> rejected	User or system removed the task, including disabled-agent auto-rejection.

2.10 Layer 3 - shared state

Shared artifact	Read by	Written by	Purpose/details
Documents	All agents and Chat Orchestrator.	Chat Orchestrator; Engineering can edit tech_notes.	Types listed: mission, product_overview, tech_notes, brand_voice, user_research.
Reports	All agents via query_reports and Chat Orchestrator.	Research, Data, Cold Outreach, Meta Ads, and any agent with create_report.	Typed reports with metadata, searchable by date and type.
Memory	Chat Orchestrator and agents as injected context.	Updated automatically or by agents across runs.	Layer 1 domain memory, Layer 3 cross-company memory, and per-agent compressed memory.

Shared artifact	Read by	Written by	Purpose/details
Context Graph	Chat via read_context_graph and agents as injected context.	Platform-maintained company knowledge.	Nodes listed for Longer include product_knowledge, revenue, active_work, support_history, user_context, feature:ads, feature:stripe_connect, feature:twitter, and similar feature-specific nodes.
Contacts and Leads	Cold Outreach and Chat indirectly via reports.	Cold Outreach.	Tracks name, email, status, and notes. Lead states: pending, contacted, replied, responded, meeting, dead.
Postgres database	Data, Engineering, and the live app.	Engineering for schema changes; live app for runtime data.	Tables/data named in the texts include users, sessions, exercises, AI usage, subscriptions, and daily_tracker.

3. Execution agents - roster

#	Agent	User-facing?	Primary purpose
29	Research	Yes	Reports, competitive analysis, market intelligence, market sizing, SEO and content strategy recommendations.
30	Engineering	Yes	Full-stack Express + Postgres web development, deployment, logs, database work, SEO, payments, and bug fixes.
32	Support	Yes	Read and reply to customer emails, escalate issues, propose engineering tasks for bugs or features.
33	Data	Yes	Database queries, business intelligence, product metrics, reports, and logs analysis.
42	Browser	Yes	Cloud browser automation, page reading, form filling, submissions on allowed sites, screenshots, and testing.
52	Meta Ads Manager	Yes	AI video ad creation, Meta campaign setup, optimization, analytics, pausing, archiving, and reports.
53	Twitter	Yes	Posts text tweets from shared @polsia account using company context and reports.
54	Cold Outreach	Yes	Find professional emails, verify them, send limited cold emails, manage lead pipeline, handle replies.
-	Chat Orchestrator	Yes	The user-facing conversation layer and task router.
-	CEO / Cycle Planner	No	Infrastructure planner that cannot be disabled by the user according to the final text.
-	Monitoring	No	Infrastructure agent mentioned as not user-controllable.
-	Reporting	No	Infrastructure agent mentioned as not user-controllable.

4. Agent #29 - Research

Property	Value
Purpose	Web research, competitive analysis, market intelligence, market sizing, pricing research, community analysis, SEO/keyword research, content strategy, and structured reports.
Max turns	200.
Model	Claude (Anthropic), as stated in the complete architecture text.
Primary output	Structured reports saved through create_report().
Critical limitation	No explicit web search or browser tools are mounted in the final architecture spec. The text says it relies heavily on model training knowledge and any runtime web search capabilities provided.

MCP mount	Tools provided
tasks	create_task_proposal, get_available_tasks, start_task, complete_task, fail_task, block_task.
reports	create_report, query_reports, get_reports_by_date.
polsia_support	report_bug, suggest_feature.

4.1 Research inputs, outputs, handoffs, failures

Category	Details
Inputs	Task description, company documents such as product overview and brand voice, prior reports, and compressed memory from prior research runs.
Required output	A report must be created before the task is completed. Report format described: Executive Summary, Key Findings with sources, and Recommended Actions.
Follow-up proposals	Can propose Engineering tasks for technical needs, Cold Outreach tasks or reports for outreach targets, and can fail with a reason when blocked by access.
Failure handling	If a source is paywalled, note the limitation and proceed with available information. If the question is too broad, narrow scope and propose a follow-up. If no useful information is found, fail with explanation.
Cannot	Cannot access paywalled academic/premium content, private Reddit or Discord communities, Google Analytics or Search Console, conduct surveys/interviews/focus groups, or guarantee recency according to the earlier verified breakdown.

5. Agent #30 - Engineering

Property	Value
Purpose	Full-stack web development: build, fix, refactor, deploy, query database, manage schema, and ship features.
Max turns	200.
Model	Claude (Anthropic).
Primary output	Code changes deployed to production or staging, with verification through status and logs.
Architecture constraint	Single Express + Postgres app on Render, not multi-service architecture.

Tool	What it does
create_instance	Spin up a new Express + Postgres app from scratch.
push_to_remote	Deploy code changes to staging or remote. The texts say Engineering must push after every file change.
push_to_prod	Deploy to production on Render.
get_status	Check app health.
get_logs	View application logs, including filtering by time or pattern in one verified text.
get_preview	Preview changes before deployment.
query_db	Run SQL directly against the Postgres database.
list_instances	See all running apps.
delete_instance	Remove an app.
create_report	Save a report.
create_task_proposal	Suggest follow-up work.

5.1 Engineering skill files

Skill file / area	Teaches or enables
ai_features.md / AI features	Polsia proxy usage for Claude and OpenAI.
frontend_ui.md / Frontend/UI	Tailwind and shadcn/ui patterns.
stripe_payments.md / Stripe payments	Checkout, subscriptions, webhooks, payment links.
neon_database.md / Neon Postgres	Postgres schema and migrations.
r2_proxy.md / R2 proxy	File uploads.
email_proxy.md / Email proxy	Sending email from app code.
openai_proxy.md / OpenAI proxy	Image generation, embeddings, OCR.
render_infrastructure.md / Render infrastructure	Deploy constraints, including 512MB RAM and no Docker.

5.2 Engineering can and cannot do

Can do	Cannot do / hard limits
- Build full-stack Express + Postgres web apps. - Write JavaScript/Node.js across backend and frontend. - Use Tailwind CSS and shadcn/ui styling patterns. - Integrate Stripe checkout, payment links, subscriptions, and webhooks. - Run database migrations and schema changes. - Add SEO elements: meta tags, Open Graph, sitemaps, robots.txt, structured data, and keyword-targeted content pages. - Build auth flows such as email signup/login and session management. - Create API endpoints, landing pages, and AI features via Polsia proxies. - Use image generation via OpenAI proxy and file uploads via R2 proxy. - Fix bugs, refactor code, run SQL queries, and view deploy logs and app health.	- Build native iOS or Android apps. - Build Python services or heavy ML frameworks such as PyTorch, TensorFlow, or Prophet. - Use multi-service architectures; the stated model is one Express app. - Connect to the user's existing GitHub repos; the texts say only Polsia-created repos. - Deploy to AWS, Vercel, Netlify, or other hosts; the stated host is Render. - Change hosting provider or access servers outside Polsia infrastructure. - Access other companies' databases. - Perform Google Search Console verification without the user's account. - Integrate arbitrary third-party APIs outside the listed supported capabilities when the text says they are not available. - Exceed 512MB RAM during builds. - Use Docker, native binaries, or desktop apps. - Build non-web native products.

5.3 Engineering execution pattern, handoffs, and failures

1. Read the task description.
2. Read relevant skill files.
3. Explore the current codebase by listing files and reading existing code.
4. Plan the changes.
5. Write or edit files.
6. Call `push_to_remote` after every file change because timeouts can lose unpushed work.
7. Wait for deployment.
8. Call `get_status` to verify the app.
9. Call `get_logs` to check for errors.
10. Fix errors and re-push if needed.
11. Call `complete_task` when verified working.

Situation	Handoff or failure behavior
Needs data not available in the database	<code>block_task()</code> explaining what is needed.
Finds a deeper bug	<code>create_task_proposal(type: bug)</code> .
CSS or design is ambiguous	Complete the functional part and propose a styling follow-up.
Major feature completed	Often proposes a test-and-verify follow-up.
Build fails due to 512MB	Remove heavy dependency or try a lighter alternative.
Deploy fails	Check logs, fix, and re-push.
Database migration fails	Rollback and try a different approach.
Runs out of 200 turns	<code>fail_task()</code> with what was accomplished.
Ambiguous requirements	Make the best judgment and document assumptions.
Conflicting skill instructions	Follow the most recent or specific skill file.

6. Agent #32 - Support

Property	Value
Purpose	Customer email support: read inbox, reply to users, maintain threads, and propose follow-up tasks.
Max turns	200.
Model	Claude (Anthropic).
Primary output	Plain text email replies sent to customers, plus reports or task proposals when useful.

MCP mount	Tools provided
company_email	get_inbox, send_company_email, get_email_thread, add_contact, get_contacts.
tasks	create_task_proposal, start_task, complete_task.
reports	create_report.

6.1 Support capabilities and limits

Can do - Read inbound emails to longer@polsia.app. - Reply to customer support questions. - Maintain conversation threads. - Escalate complex issues. - Create Engineering task proposals for technical issues users report. - Add contacts and produce support reports such as volume or common questions.	Cannot do - Access the user's personal email. - Make phone calls. - Live chat with users in real time. - Access help desk platforms such as Zendesk or Intercom. - Promise features that do not exist. - Share sensitive technical details such as API keys or database information.
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6.2 Support rules, handoffs, and failures

Situation	Behavior
Technical bug reported by user	create_task_proposal(tag: engineering, type: bug).
Feature request from user	create_task_proposal(tag: engineering, type: feature).
Billing or subscription issue	Respond with guidance and escalate if needed.
Angry or sensitive user	Respond empathetically and flag for human review.
Question cannot be answered from docs	Reply honestly that the issue will be looked into.
Email send fails	fail_task() with error.
Spam or irrelevant email	Ignore and complete the task.

- Email format must be plain text only, with no HTML and no markdown rendering.
- Response length should match question complexity.
- The signature should be the company name, not AI or bot.

7. Agent #33 - Data

Property	Value
Purpose	Business intelligence: query database, analyze metrics, inspect logs, and generate reports.
Max turns	200.
Model	Claude (Anthropic).
Primary output	SQL query results and reports.
Important role note	The final spec says Data has a full polsia_infra mount and technically can push code, but its role instructions focus it on analysis and reporting, not feature building.

Tool	What it does
query_db	Run arbitrary SQL against Postgres.
get_logs	View application logs.
get_status	Check application status.
create_instance, push_to_remote, push_to_prod, get_preview, list_instances, delete_instance	Available through polsia_infra, but not the intended focus for Data.
create_report, query_reports	Save and read analysis reports.
create_task_proposal, start_task, complete_task	Manage its task lifecycle and propose follow-ups.

7.1 Data can and cannot do

Can do - Run SELECT, INSERT, and UPDATE queries according to one source text. - Count users and track signups over time. - Measure engagement and AI message usage patterns. - Check subscription conversion rates. - Generate cohort analyses. - Build custom metrics dashboards as reports. - Check app logs for patterns.	Cannot do / limits - Connect to external databases or APIs. - Access Google Analytics, Mixpanel, or Amplitude. - Build real-time live dashboards; reports are point-in-time unless Engineering builds tracking. - Access data outside the app's Postgres instance.
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7.2 Data execution pattern and failure behavior

1. Read the task, such as a user-count or sign-up analysis question.
2. Discover schema using information_schema queries.
3. Explore relevant table columns.
4. Write and run SQL queries.
5. Analyze the result.
6. Create a report with findings.
7. Complete the task.

Situation	Behavior
Data reveals a bug such as orphaned records	Propose Engineering bug task.
Analysis suggests product change	Propose Engineering task.
Data is insufficient	Report what is available and propose logging or tracking improvements.
Table does not exist	Report limitation and suggest Engineering add tracking.
Query times out	Simplify, use LIMIT, or try a different approach.
Metric definition is ambiguous	Make the assumption explicit in the report.

8. Agent #42 - Browser

Property	Value
Purpose	Automated web interaction: public page browsing, scraping/extraction, form filling, submissions, screenshots, and end-to-end product testing.
Max turns	Not explicitly capped in one text, but session timeout is about 20 minutes.
Model	Claude (Anthropic).
Primary output	Completed web actions, screenshots, extracted data, and task completion or failure reports.

Property	Value
Critical constraint	Site tier enforcement is hardcoded and cannot be overridden according to the complete architecture text.

Tool	What it does
browserbase_session_create	Open a cloud browser session.
browserbase_navigate	Go to a URL.
browserbase_screenshot	Take a screenshot.
browserbase_click	Click an element or CSS selector.
browserbase_fill	Type into a form field.
browserbase_extract	Extract structured data from a page.
browserbase_get_page_content	Get full page HTML/text.
browserbase_evaluate	Run JavaScript on the page.
browserbase_session_close	Close the browser.
get_site_tier	Check the allowed access tier for a website.
get_or_create_browser_context	Use or create persistent login context across runs.
save_site_credentials, get_site_credentials	Store or retrieve saved logins.
check_verification_inbox	Read verification emails for account creation.
get_inbox, send_company_email	Email support for verification or related tasks.
create_task_proposal, start_task, complete_task	Task lifecycle and follow-up proposals.

8.1 Browser site tiers

Tier	Examples from the texts	Login	Post	Create account	Notes
1	Reddit, Twitter/X, Instagram, LinkedIn, TikTok, Facebook, ProductHunt, IndieHackers, YouTube	No	No	No	Browse/read only. Cannot DM, post, create accounts, or log in.
1.5	HackerNews, Medium, Dev.to, Gumroad, Etsy, Craigslist, Wellfound	Existing credentials only	Yes	No	New account creation is blocked by CAPTCHA according to the texts.
2	Hashnode, Substack, BetaList, Lobste.rs, smaller forums	Yes	Yes	Yes	Full access: create accounts, log in, post, and interact.
3	Everything else	Yes	Yes	Yes if no CAPTCHA	Browse by default and can create accounts when needed if not blocked.

8.2 Browser can and cannot do

Can do - Browse any public webpage. - Read Reddit threads, profiles, and comments, without interacting. - Fill forms and create accounts on Tier 2/3 sites using longer@polsia.app when allowed. - Submit the app to allowed directories such as BetaList. - Post on smaller allowed communities such as Hashnode, Substack, BetaList, and Lobste.rs where tier rules permit. - Test the live app end-to-end, including signup flow and payments. - Scrape public page content in a single browser-session manner. - Take screenshots at key steps. - Maintain persistent login sessions across runs when credentials are available and tier allows it.	Cannot do - Post on Reddit, Twitter/X, Instagram, LinkedIn, ProductHunt, IndieHackers, TikTok, Facebook, or YouTube. - DM anyone on Tier 1 platforms. - Create Reddit, Twitter/X, Instagram, LinkedIn, TikTok, ProductHunt, or IndieHackers accounts. - Bypass CAPTCHAs on Tier 1.5 sites. - Access sites behind 2FA that has not been set up. - Download large files. - Run massive-scale scraping; it is one browser session per task. - Interact with desktop apps. - Run beyond the stated session cap of about 20 minutes per task.
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8.3 Browser execution, handoffs, and failure behavior

- 1. Read the task, such as a site submission or product test.
- 2. Call `get_site_tier(site)` to determine allowed actions.
- 3. If the site is Tier 1, browse/read only and do not interact.
- 4. If the site is Tier 2 or higher, proceed with allowed interaction.
- 5. Load saved browser context when available.
- 6. Create the browser session.
- 7. Navigate, screenshot, click, fill, and screenshot at key steps.
- 8. Use `longer@polsia.app` for allowed account creation when needed.
- 9. Use `check_verification_inbox` when an email verification step is needed.
- 10. Complete the task with a final screenshot as proof when successful.

Situation	Behavior
Tier 1 site requested for interaction	<code>fail_task()</code> with site is browse-only.
CAPTCHA encountered on Tier 1.5	<code>fail_task()</code> with CAPTCHA blocked account creation.
Scraped data needs analysis	Propose Research or Data task.
Broken link found on user's site	Propose Engineering bug task.
Site blocks automation	<code>fail_task()</code> and suggest manual approach.
Session timeout	Use screenshots to show progress and fail with partial results.
Element not found	Try alternative selector and use screenshot for debugging.
Site layout changed	Adapt using page content or fail.
Login expired	Re-authenticate from saved credentials where possible.

9. Agent #52 - Meta Ads Manager

Property	Value
Purpose	Create AI video ads and manage Facebook/Instagram campaigns.
Max turns	100.
Model	Claude (Anthropic).
Primary output	Running ad campaigns, optimized ad sets, and performance reports.
Autonomy	Described as the most autonomous agent in one source text.

Tool	What it does
<code>create_ad</code>	Create an individual ad unit. In the first-run workflow, this one call handles Sora 2 video generation, captions, Meta upload, creative setup, ad creation, and activation.
<code>get_ad_analytics</code>	Pull analytics for existing ads.
<code>pause_ad</code> , <code>activate_ad</code> , <code>archive_ad</code>	Manage ad status.
<code>create_campaign</code>	Set up a new ad campaign.
<code>create_adset</code>	Define targeting and budget.
<code>upload_ad_video</code>	Upload video creative.
<code>create_video_creative</code>	Generate AI video ads with Sora 2.
<code>activate_campaign</code>	Start running ads.
<code>save_ad</code>	Save ad configuration.
<code>update_ad_metrics</code>	Pull or update performance data.

Tool	What it does
add_captions	Add captions to video ads.
get_ad_account	Check ad account status.
list_campaigns	See all campaigns.
get_campaign_insights	Performance analytics.
create_task_proposal, start_task, complete_task, create_report	Task lifecycle and report output.

9.1 Meta Ads can and cannot do

Can do - Generate AI video ads from scratch using Sora 2. - Create vertical iPhone selfie UGC-style ads with natural dialogue according to the first-run template. - Target by demographics, interests, and behaviors on Facebook/Instagram. - Set daily budgets in the stated \$10-\$1000/day range. - Monitor impressions, clicks, CTR, CPC, and campaign insights. - Pause underperforming ads automatically. - Create new creative variations and add captions. - Run fully managed campaigns with no user setup needed according to the verified breakdown.	Cannot do / limits - Run Google Ads, TikTok Ads, YouTube Ads, or other non-Meta ad platforms. - Target by uploaded custom audience lists. - Spend more than \$1000/day. - Guarantee a specific CPA or ROAS. - Guarantee ads in restricted or policy-sensitive categories will pass review; men's health may trigger review according to the source text.
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9.2 Meta Ads operating modes

Mode	Workflow
First run: no campaigns exist	Read company context, generate a UGC video prompt using vertical iPhone selfie style and natural dialogue, call create_ad with prompt/headline/body_text, wait 5-8 minutes per ad, verify analytics, create a launch report, and complete the task.
Optimization run: campaigns exist	Pull analytics, check disapproved or flagged ads first, evaluate each active ad against CTR/CPC/impression rules, create replacement ads before pausing old ads, produce a decision report, and complete. No action needed is valid.

Performance condition	Action
Healthy: CTR > 1% and CPC < \$1	Keep running.
Mediocre: CTR 0.5-1% or CPC \$1-2	Wait 7 days before replacing.
Underperforming: CTR < 0.5% or CPC > \$2 after 7+ days	Replace.
Delivery-failed: active 5+ days with <200 impressions	Replace.
Replacement needed	Create the new ad first, then pause the bad one so delivery does not drop to zero.

Daily budget	Max active ads	Max creations per 24h
\$10/day or less	2	2
\$10-30/day	3	3
\$30+/day	5	5

9.3 Meta Ads handoffs and failures

Situation	Behavior
All ads disapproved	Propose a new creative angle.
Landing page conversion is bad	Propose Engineering task for landing page improvements.

Situation	Behavior
Budget exhausted	Report and wait for next scheduled run.
Sora 2 generation fails	Retry with simplified prompt.
Meta rejects ad for policy	Note reason and try a different creative angle.
Ad account restricted	fail_task() and flag for support.
All ads underperform	Replace 1-2 rather than nuking everything at once.
0 impressions after 5 days	Treat as likely delivery issue and recreate campaign.

10. Agent #53 - Twitter

Property	Value
Purpose	Post tweets from the shared @polsia account.
Max turns	200.
Model	Claude (Anthropic).
Primary output	Published text tweet plus a report logging what was posted.
Important limit	Tweets are from @polsia, not the user's personal account or a dedicated Longer account.

MCP mount	Tools provided
twitter	post_tweet, get_twitter_account.
tasks	create_task_proposal, start_task, complete_task.
reports	create_report, query_reports.
documents	get_company_documents, get_company_document.

10.1 Twitter rules and limits

Can do - Post a tweet from the shared @polsia account. - Read own tweet history or get account info according to the verified breakdown. - Compose context-aware tweets based on company documents and recent reports. - Include the website link, which the detailed spec says every tweet must include.	Cannot do - Post from the user's personal Twitter/X account or a dedicated Longer account. - Send DMs. - Reply to other people's tweets. - Post threads or multi-tweet sequences. - Schedule tweets for specific times. - Read or search other people's tweets. - Post images or video.
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- Tweet voice described in the source: dark humor, witty, bitter over excited.
- No emojis and no hashtags are allowed in the detailed Twitter prompt description.
- The tweet must stay within 280 characters.
- Rate limit is described as 2/day shared across all Polsia companies in the final architecture text and as 1/day or effectively 1/day for the user in other verified text. The operational takeaway preserved here is: limited, shared platform capacity, not a dedicated distribution channel.

Failure	Response
Rate limited	fail_task() with daily limit reached.
Tweet rejected	Try a shorter version.

11. Agent #54 - Cold Outreach

Property	Value
Purpose	Find professional emails, verify deliverability, send cold outreach, manage lead pipeline, handle replies.
Max turns	200.
Model	Claude (Anthropic).
Primary output	Emails sent, lead statuses updated, inbound replies handled, and reports created.
Critical constraint	Hunter.io needs a real name and company domain. It cannot look up anonymous Reddit usernames alone.

MCP mount	Tools provided
company_email	get_inbox, send_company_email, get_email_thread, add_contact, get_contacts.
hunter_io	find_email, verify_email.
tasks	create_task_proposal, start_task, complete_task.
reports	create_report.
documents	get_company_documents.

11.1 Outreach can and cannot do

Can do

- Find a person's professional email when name and company domain are known.
- Verify an email before sending.
- Send short personalized cold emails from longer@polsia.app.
- Track lead status over time.
- Send follow-ups after 5+ days when there is no reply.
- Reply to inbound responses without counting against the cold email cap.
- Add contacts and maintain a lead pipeline.

Cannot do

- Find personal Gmail, Yahoo, Hotmail, or similar addresses.
- Send more than 2 cold emails per day to strangers.
- Send from a custom domain; the source states longer@polsia.app only.
- Send HTML-formatted emails or attach files according to the verified breakdown.
- Guarantee inbox delivery.
- Send bulk blasts.
- Find emails from Reddit usernames alone.
- DM people on any social platform.

11.2 Outreach workflow, status machine, and limits

1. Check replies first with get_inbox and respond to any lead replies. Replies are unlimited.
2. Research new leads if the pipeline is below threshold. Use find_email(name, domain), verify_email(), and add_contact().
3. Send cold outreach, capped at 2 cold emails per day. Verify deliverability before sending.
4. Write cold emails as 50-125 words, plain text, founder-to-founder voice, one clear CTA, and subject lines of 3-5 lowercase words with no clickbait.
5. Follow up after 5+ days if there is no reply. Max 2 follow-ups per lead. If no reply after 2 follow-ups, mark dead.
6. Create a report summarizing actions and complete the task.

Category	Limit
Cold emails to strangers	2/day.
Known contact emails	20/day.
Replies to inbound	Unlimited.
Emails to company owner	Unlimited.
51 leads math	At 2/day, 51 leads take about 25+ days, and only if real names and company domains exist.

Lead state	Meaning / transition
pending	Lead exists but has not been contacted.
contacted	Cold email sent.

Lead state	Meaning / transition
replied	Lead responded.
responded	Agent has responded to the lead's reply.
meeting	Lead reached meeting stage.
dead	Lead is no longer active, including bounce or no reply after two follow-ups.

11.3 Outreach handoffs and failures

Situation	Behavior
Lead replies with technical question	Propose Support task.
Lead asks for a feature that does not exist	Note it in a report for Chat Orchestrator to see.
Cannot find email	Report failure and move to the next lead.
Lead converts	Propose Support task for onboarding.
Hunter.io cannot find email	Skip lead and move to next.
Email bounces	Mark lead as dead and do not retry.
No leads in pipeline	fail_task() or create a report explaining the gap.
Rate limit hit	Complete what is possible and report remaining work.

12. Cross-cutting systems

12.1 Agent execution model

When a task runs, the agent receives its permanent system prompt, the task description, company context from documents and context graph, compressed memory from previous runs, and its allowed MCP tools. It then runs autonomously, making tool calls and deciding next steps until it completes, blocks, or fails.

- Agents are isolated AI sessions, not live chats with the user.
- Agents do not communicate directly with each other.
- Agents can create task proposals, reports, code, database changes, emails, ads, screenshots, or other outputs depending on their tool mounts.
- Coordination happens through task queue order, shared reports, documents, database state, memory, context graph, recurring schedules, and planner outputs.

12.2 Coordination example from the source texts

1. Research produces a report identifying 51 active Reddit users in PE communities.
2. Chat Orchestrator sees the report and creates a Cold Outreach task to email those leads.
3. Cold Outreach reads the report and tries to find emails via Hunter.io.
4. If emails are missing, Cold Outreach proposes Browser work to check profiles for public contact info.
5. Browser can browse Reddit profiles because Reddit is Tier 1 read-only, but cannot interact.
6. Any visible emails or websites go back into reports.
7. Cold Outreach can then send only if valid professional emails are found and rate limits allow.

12.3 Shared tool layer

Tool	Purpose
create_task_proposal	Suggest follow-up work.
get_available_tasks	See other tasks in queue.

Tool	Purpose
start_task, complete_task, fail_task, block_task	Manage execution lifecycle.
create_report, query_reports	Read and write business reports.
report_bug, suggest_feature	Flag platform issues or product suggestions to Polsia.

12.4 Memory architecture

Memory layer	Scope	Purpose	Read by
Layer 1: Company Domain Knowledge	This company; text states 15K tokens.	Accumulated facts about Longer, decisions made, preferences expressed, and technical details discovered. Updated automatically as agents learn.	Chat Orchestrator and all agents as context.
Layer 3: Cross-Company Patterns	Platform-wide; text states 15K tokens.	Patterns learned across all Polsia companies, what works for similar tasks, and common failure modes.	Chat Orchestrator.
Per-agent compressed memory	Each agent.	Notes from prior runs, injected into context at execution time to reduce repeated mistakes.	Same agent on the next run.

12.5 Learnings and skills systems

System	What the text says
Learnings	A tagged knowledge base separate from memory. Agents create learnings when they discover something useful. Learnings are searchable by tag and persist across runs. Example given: Reddit is Tier 1, cannot post, tagged browser and reddit.
Skills	Files in .claude/skills/ that Engineering reads. They teach app-specific patterns and are updated when the platform adds capabilities. Other agents do not use skill files because their roles are simpler.

12.6 Platform services available outside a single agent

Service	Mount	What it provides
Postmark	postmark	Raw transactional email sending.
Scripts	scripts	Run, list, and get output from scripts.
Cycle Planning	cycle_planning	Autonomous daily planning cycles.
Agent Factory	agent_factory	Create new custom agents with specific tool combinations.
Learnings	learnings	Persistent knowledge base: create, search, and tag learnings.
Dashboard	dashboard	Company dashboard management.

12.7 What is opaque from the source perspective

Component	What is visible in the texts	What is not visible
Queue dispatcher	Tasks route by tag to agent, or by suggested_agent_id.	Exact matching algorithm, retry logic, and scheduling priority.
Credit system	0 credits blocks execution.	Whether credits are consumed per task, per turn, per tool call, or another formula.
Cycle planner	Exists and runs daily.	Planning algorithm, scoring model, and feature flags.
Boost planner	Exists and proposes tasks.	Trigger conditions and frequency.
Model routing	Claude (Anthropic) is stated.	Exact model version, temperature, and token limits per agent.
Agent Factory	Can create custom agents.	Full API and exact allowed tool combinations.
Scripts system	Exists as an MCP mount.	Which scripts exist and how to run them.
Rate limiting	Some hard limits are named, including 2 cold emails/day and Twitter limits.	All platform-level throttles.
Monitoring	Mentioned as infrastructure.	What it monitors and alert conditions.

Component	What is visible in the texts	What is not visible
Reporting agent	Mentioned as infrastructure.	What triggers it and how automated report generation works.

13. Direct Chat Polsia capabilities without task credits

The verified breakdown states that certain actions happen directly in conversation without agent execution or task credits.

Action	Tool or mechanism
Answer questions about the company	Context graph and memory.
Draft email copy, landing page copy, and content	Chat generation for the user to copy-paste.
Create, edit, reject, and reorder tasks	Task management tools.
Read or update company documents	Document tools for mission, brand voice, tech notes, product overview, and user research.
Check what is running	Active executions.
View task logs	Execution logs.
Manage recurring tasks	Create, update, or delete schedules.
Rename company	rename_company.
Manage dashboard links	Link tools.
Pause Meta Ads immediately	pause_ads.
Disable or enable agents	Agent management.
Open support tickets to Polsia team	open_support_ticket.
Search memory and knowledge	Memory tools.
View reports	Report queries.
Read emails	get_emails.
Read tweets	get_tweets.

14. Capability boundaries by domain

14.1 Engineering and app infrastructure

- Build, fix, and deploy full-stack Express + Postgres web apps on Render.
- Use Stripe, Tailwind, shadcn/ui, Polsia AI proxies, R2 proxy, email proxy, SEO files, Open Graph, sitemaps, robots.txt, structured data, auth, sessions, APIs, landing pages, database migrations, SQL, and logs.
- Cannot build native mobile apps, connect to arbitrary existing GitHub repos, deploy to non-Render hosts, use Docker, exceed the 512MB build limit, or implement heavy ML services.

14.2 Browser and external sites

- Can browse public pages and interact only where site tiers allow.
- Can submit to Tier 2 sites such as BetaList, Hashnode, Substack, Lobste.rs, and smaller forums when account creation and posting are allowed.
- Cannot post or DM on Tier 1 platforms, including Reddit and major social platforms.

14.3 Outreach and communications

- Cold Outreach can send from longer@polsia.app, not a custom domain in the source text.
- Cold outreach is intentionally slow: 2 cold emails/day to strangers, 20/day to known contacts, unlimited replies.
- Support can handle inbound customer email but not the user's personal inbox, phone, or live chat.

14.4 Marketing and ads

- Meta Ads can generate AI video creative, upload, caption, activate, monitor, pause, and replace ads on Facebook/Instagram.
- Meta Ads cannot run Google, TikTok, or YouTube ads and cannot guarantee CPA or ROAS.
- Twitter can post limited text tweets from @polsia only, with no replies, DMs, threads, images, video, or dedicated brand account posting.

14.5 Research, data, and reporting

- Research produces reports and can cover competitive analysis, pricing research, market sizing, SEO/keyword research, community analysis, and content strategy.
- Research has limited explicit tools and should not be treated as a live web-browsing agent based on the later source texts.
- Data queries Postgres and logs to produce BI reports, but cannot connect to external analytics databases or third-party analytics tools.

14.6 Content and creative outputs

Can do	Cannot do
- Write blog posts and landing page copy. - Write SEO articles, email templates, and newsletter drafts. - Create brand voice documentation. - Produce AI-generated ad creative through the Meta Ads agent flow where supported.	- Produce non-ad video content outside the AI ad flow described in the texts. - Record podcasts or audio. - Design graphics through Figma or Canva access. - Provide photography.

15. Longer-specific implications from the text files

15.1 What works for Longer right now

Goal	Best path described	Realism described in the texts
Get 51 Reddit leads to the app	User personally DMs them on Reddit.	Strongest channel because Polsia cannot DM Reddit users.
Cold email outreach	Cold Outreach agent, 2/day, if real name and company domain are known.	Slow and requires real identities/domains.
SEO traffic	Engineering builds meta tags, sitemap, robots.txt, structured data, and more keyword-targeted content pages.	Works well according to the verified breakdown.
Paid traffic	Meta Ads with at least the stated minimum budget context of \$10/day in the source text.	Fully managed within Meta constraints.
Community posting	Browser posts or submits only on Tier 2/Tier 3 sites such as Hashnode, Substack, BetaList, or Lobste.rs.	Works where tier rules allow.
Twitter presence	Shared @polsia account posts.	Limited value because it is not a dedicated brand account and rate-limited.
Test product	Browser runs full end-to-end test.	Works within session time and site constraints.
Track user behavior	Data agent queries Postgres.	Works for app database data.

15.2 Distribution bottleneck

The source texts repeatedly identify distribution as the bottleneck for Longer. The strongest channel named is the user's own Reddit account for personal DMs or posts, because Polsia agents can research Reddit but cannot interact there.

- SEO foundation is repeatedly named as a high-leverage agent-executable next step.
- BetaList submission is named as an agent-executable distribution path because BetaList appears in Tier 2 examples.
- Product polish, onboarding flow, and mobile UX are named as valuable Engineering focuses so visitors convert once they arrive.
- Meta Ads are a fully managed paid option, subject to budget, policy review, and performance thresholds.
- The text states that 0 credits blocks execution, so queued work cannot run until credits exist.

16. Corrections and superseded claims

Previously implied or overstated	Corrected version kept in this document
Browser might post on Reddit/forums if it creates an account and logs in.	Reddit is Tier 1 browse-only. Browser can read Reddit but cannot log in, create accounts, post, or DM.
Reddit leads were actionable leads.	They are research leads unless public contact information can be found. Without public emails or real names plus company domains, they are not actionable through Polsia outreach.
Browser can post on ProductHunt or IndieHackers.	ProductHunt and IndieHackers are Tier 1 browse-only in the later texts.
Research agent has strong explicit web search capabilities.	Research has tasks, reports, and support tools. It relies on model knowledge and any runtime search support, with create_report as its required output.
Twitter can build meaningful brand presence through the user's account.	Twitter posts from the shared @polsia account, not the user's personal or Longer account, and cannot reply, DM, thread, or post media.
Cold outreach scale can cover 51 leads quickly.	At 2 cold emails/day, 51 leads take about 25+ days and only if viable professional emails can be found.
Browser automation is unlimited browsing or scalable scraping.	It is a single browser session per task with about a 20-minute cap.
Earlier summaries omitted Agent Factory, Scripts, Cycle Planning, Learnings, Dashboard, Monitoring, and Reporting infrastructure.	Those are included here only as described in the later text files.

17. Exhaustive source checklist

This checklist shows how the repeated source content was folded into the consolidated structure without duplicating entire paragraphs.

Source content area	Where it appears in this PDF
Original transparent breakdown and Reddit lead mistake	Control points, corrections, and Longer-specific implications.
Verified eight-agent capability breakdown	Agent specs, tool lists, can/cannot lists, practical assessment, direct Chat capabilities, platform services.
Top-down architecture overview	System architecture, task creation, agent execution model, coordination, shared tool layer.
Complete architecture spec	Layer model, Chat Orchestrator tools, planners, queue schema, dispatch gates, shared state, complete agent specifications, cross-cutting memory/skills/learnings, opaque systems, full roster.

17.1 Explicit non-inclusions

- No information from public web search or outside sources is included.
- No claims from generated PDFs were used unless the same content existed in the four text files.
- No obsolete claim is presented as an active capability when a later text corrected it.

18. Source files

The PDF was generated from these four text files only:

- Pasted text(17).txt - original transparent breakdown and first capability list.
- Pasted text(18).txt - corrected verified eight-agent breakdown.
- Pasted text(21).txt - top-down architecture and agent execution overview.
- Pasted text(22).txt - complete system architecture, layer model, full agent specs, and cross-cutting systems.

End of consolidated document.